Waseem Raza, PhD (Electrical and Computer Engineering)

ECE Lecturer | Communication Networks, Programming & Applied Machine Learning

Undergraduate & Graduate Instruction | Wireless Networks | Data-Driven Engineering

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LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

Academic Profile

Electrical and Computer Engineering PhD (University of Oklahoma, 2025) with prior university-level teaching in Computer Programming, Communication Systems, and Wireless Communication. Experienced in engineering laboratory support, student-project mentoring, curriculum-aligned assignments and assessment. Doctoral research in AI/ML for wireless networks complements industrial research and engineering experience at AT&T Labs, Futurewei, and AT&T. Prepared to contribute to Rutgers ECE lectures, discussions and applied coursework aligned with communication networks, wireless systems, programming, and machine learning for engineers.

Research Interests

Computer and Communication Networks; Wireless and Mobile Communication Systems; Machine Learning for Engineers and Networked Systems; Data-Driven Signal Analysis and Network Optimization; Reliability, Robustness and Engineering Experimentation

Doctoral Research Focus

Dissertation on improving resilience and accuracy of AI models and wireless networks through outage detection, data-driven optimization, generative propagation modeling and robustness under distribution shift.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*

University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*

University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*

University of Engineering and Technology, Taxila, Pakistan

Relevant Graduate Coursework

LTE Networks and Telecommunications; Emerging 5G Systems; Advanced ML and Computer Vision; Wireless Testbed Design; Numerical Analysis; Information Theory; Digital Signal Processing

Teaching, Research & Professional Experience

Teaching & Instruction

Lecturer, Department of Technology **12/2017 - 09/2019**

University of Lahore, Pakistan *Lahore, Pakistan*

- Taught programming, communication systems, and wireless communication with practical examples.
- Advised senior-design students on projects, academic plans, and early careers.
- Aligned course materials with learning outcomes and accreditation standards.
- Supported accreditation, exams, technical workshops, and student activities.

Teaching areas: Computer Programming; Communication Systems; Wireless Communication; Computer and Communication Networks (related technical expertise); Machine Learning for Engineers (related research and industry expertise); Communication Laboratories and Applied Computational Projects

Student mentoring and research development

- Advised undergraduate student projects in communications, programming and related engineering topics.
- Supported graduate and undergraduate research students on simulation, experimental design and technical reports.
- Developed laboratory exercises and practical examples connecting communication theory with implementation.

University Research Appointments

Graduate Research Assistant **09/2019 - 07/2025**

AI4Networks, University of Oklahoma *Tulsa, OK*

- Built 3GPP-compliant Python simulation tools and an AI-enabled 5G research testbed.
- Built GAN, transfer-learning, autoencoder, and reinforcement-learning models for wireless data.
- Led 100+ experiments testing wireless ML under domain shift and limited labeled data.
- Developed self-healing and outage-management methods, improving resilience by 30%.
- Published wireless AI research and mentored junior researchers in experimental design.

Research Associate, ComSens Lab **06/2016 - 12/2017**

COMSATS University Islamabad, Pakistan *Islamabad, Pakistan*

- Modeled wireless communication systems through simulation, analysis, and technical reports.
- Mentored Bachelor's and Master's theses in experiment design and research presentation.
- Reviewed student reports and manuscripts for rigor and publication readiness.
- Supported proposals, literature reviews, lab seminars, and research coordination.

Full-Time Research Scholar, Communication Systems Lab **08/2014 - 02/2016**
University of Engineering and Technology, Taxila, Pakistan *Taxila, Pakistan*

- Researched wireless communication systems while completing an MSc in Telecommunications.
- Supported communications courses, labs, assignments, grading, and examinations.
- Led wireless communication labs and prepared technical manuals and experiments.
- Developed course and student-project materials for practical wireless engineering.

Industry Research & Engineering

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Applied domain-adaptive ML to telecom service assurance and model validation.
- Designed cross-domain evaluation and error-analysis methods for operational ML.
- Worked with engineering and public-safety teams on reliable network assurance.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built LLM-enabled analytics to interpret geochemical well data for decision support.
- Built Python pipelines in Azure Databricks, Functions, and Synapse Analytics.
- Automated exploration-data ETL, cutting preparation time by 40%.
- Used data-validation and quality controls to improve output reliability by 20%.
- Turned geologists' technical needs into structured datasets and useful analytics.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR logs with ML, reducing diagnostic time by 20%.
- Improved 5G NR-NTN cell selection by 15% through data-driven validation.
- Presented operational data insights to next-generation wireless engineering teams.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved wireless CSI compression by 25% with transfer learning and autoencoders.
- Ran 50+ experiments on wireless ML robustness under changing data conditions.
- Translated research results into recommendations for future wireless networks.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Improved 5G standalone user-timing precision by 20% with ML and statistics.
- Analyzed user traces with SQL and presented results to engineering leadership.
- Turned operational traces into indicators for network diagnosis and optimization.

Research Contributions

Publications

AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks *Computer Communications* 2025 | DOI

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *Sensors 2025* 2025 | DOI

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *IEEE PIMRC* 2024 | DOI

Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity *IEEE VTC-Fall 2024* 2024 | Link

An AI-Enabled Framework to Defend Ingenious MDT-Based Attacks on the Emerging Zero Touch Cellular Networks *IEEE Network, 38(1), 228-237* 2024 | DOI

Positioning Error Impact Compensation through Data-Driven Optimization in User-Centric Networks *IEEE GLOBECOM 2023* 2023 | Link

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) *IEEE Network* 2022 | Link

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *IEEE WCNC* 2022

Research Projects & Platforms

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

TurboRAN Experimental Platform *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

Tools & Methods: 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

Research Methods & Technical Expertise

Documented Teaching: Undergraduate Computer Programming, Communication Systems and Wireless Communication; assignments, laboratory support, student guidance and assessment

Computer & Wireless Networking: Network protocols and performance, LTE/4G, 5G-NR, radio-access networks, wireless propagation, network measurements, reliability and data-driven optimization

AI/ML & Data Analysis: Supervised learning, classification, regression, anomaly detection, generative modeling, transfer learning, reinforcement learning, data evaluation and visualization

Programming & Tools: Python, C/C++, MATLAB, SQL, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Git and Linux

Teaching Practice: Worked problems, reproducible simulations, formative feedback, accessible assignments, structured office hours and project-based learning

Academic Service & Recognition

Academic Service

- Supported engineering assessment, examinations and curriculum-related activities.
- Contributed to laboratory instruction, technical seminars and student mentoring.

Honors & Awards

William H. Barkow Scholarship *2024*
University of Oklahoma, Electrical and Computer Engineering

Clyde L. Farrar Endowment Scholarship *2024*
University of Oklahoma

P.H. Robinson Fellowship *2023*
University of Oklahoma

FAS-Charles Forgan Scholarship *2023*
University of Oklahoma

Nettie Vincent Boggs Graduate Endowment *2022*
University of Oklahoma

SHS-Paul D. Massad Scholarship *2022*
University of Oklahoma

Cleo Cross Memorial Scholarship *2021*
University of Oklahoma

Marjorie Haskell Engineering Scholarship *2021*
University of Oklahoma

Fully Funded PhD / Graduate Research Assistantship *2019*
University of Oklahoma

Self-Learning

1. Machine Learning Stanford/Coursera | Completed | Mar 2020
Supervised and unsupervised learning, regularization, model evaluation, and practical predictive modeling.

2. Python for Data Science, AI & Development IBM/Coursera | Completed | Jul 2026
Python programming, data analysis, APIs, and reproducible data-science workflows.

3. AI Foundations & Industry Overview for Telecommunication Coursera | Completed | Jun 2026
AI applications in telecom networks, operations analytics, service assurance, and automation.

4. Network and Security Optimization in Telecommunication Coursera | Completed | Jun 2026
Data-driven network optimization, telecom security, reliability analysis, and operational efficiency.

5. 4G Network Fundamentals Mines-Télécom/Coursera | Completed | Aug 2026
LTE radio and core networks, cellular architecture, mobility management, and 4G fundamentals.